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Compare Permissions on Regular Files and Directories

setup

```
(prathiksha@kali)~  
$ cd ~  
  
(prathiksha@kali)~  
$ mkdir permission_test  
  
(prathiksha@kali)~  
$ cd permission_test  
  
(prathiksha@kali)~/permission_test  
$ pwd  
/home/prathiksha/permission_test  
  
(prathiksha@kali)~/permission_test  
$ echo "hello world" > sample.txt  
  
(prathiksha@kali)~/permission_test  
$ mkdir sample_dir  
  
(prathiksha@kali)~/permission_test  
$ echo "inner content" > sample_dir/inner.txt  
  
(prathiksha@kali)~/permission_test  
$ ls -l  
total 8  
drwxrwxr-x 2 prathiksha prathiksha 4096 Aug 22 18:11 sample_dir  
-rw-rw-r-- 1 prathiksha prathiksha 12 Aug 22 18:10 sample.txt
```

Testing 400 (r--)

```
(prathiksha@kali)~/permission_test  
$ chmod 400 sample.txt sample_dir  
  
(prathiksha@kali)~/permission_test  
$ ls -l  
total 8  
dr----- 2 prathiksha prathiksha 4096 Aug 22 18:11 sample_dir  
-r----- 1 prathiksha prathiksha 12 Aug 22 18:10 sample.txt  
  
(prathiksha@kali)~/permission_test  
$ cat sample.txt  
hello world  
  
(prathiksha@kali)~/permission_test  
$ echo "test" >> sample.txt  
zsh: permission denied: sample.txt  
  
(prathiksha@kali)~/permission_test  
$ ls sample_dir  
ls: cannot access 'sample_dir/inner.txt': Permission denied  
inner.txt  
  
(prathiksha@kali)~/permission_test  
$ cd sample_dir  
cd: permission denied: sample_dir
```

Read succeeded, modify was denied on the file. Listing the directory succeeded, but entering it was denied.

Testing 200 (-w-)

```
(prathiksha@kali)~/permission_test  
$ chmod 700 sample.txt sample_dir  
  
(prathiksha@kali)~/permission_test  
$ chmod 200 sample.txt sample_dir  
  
(prathiksha@kali)~/permission_test  
$ ls -l  
total 8  
d-w----- 2 prathiksha prathiksha 4096 Aug 22 18:11 sample_dir  
-w----- 1 prathiksha prathiksha 12 Aug 22 18:10 sample.txt  
  
(prathiksha@kali)~/permission_test  
$ cat sample.txt  
cat: sample.txt: Permission denied  
  
(prathiksha@kali)~/permission_test  
$ echo "test2" >> sample.txt  
  
(prathiksha@kali)~/permission_test  
$ ls sample_dir  
ls: cannot open directory 'sample_dir': Permission denied  
  
(prathiksha@kali)~/permission_test  
$ cd sample_dir  
cd: permission denied: sample_dir  
  
(prathiksha@kali)~/permission_test  
$ touch sample_dir/newfile.txt  
touch: cannot touch 'sample_dir/newfile.txt': Permission denied
```

Read was denied, modify succeeded on the file. Listing, entering, and creating a file in the directory were all denied

Testing 100 (--x)

```
(prathiksha@kali)~[/permission_test]
$ chmod 700 sample.txt sample_dir

(prathiksha@kali)~[/permission_test]
$ chmod 100 sample.txt sample_dir

(prathiksha@kali)~[/permission_test]
$ ls -l
total 8
d--x----- 2 prathiksha prathiksha 4096 Aug 22 18:11 sample_dir
--x----- 1 prathiksha prathiksha 18 Aug 22 18:16 sample.txt

(prathiksha@kali)~[/permission_test]
$ cd sample.txt
cd: not a directory: sample.txt

(prathiksha@kali)~[/permission_test]
$ ./sample.txt
zsh: permission denied: ./sample.txt

(prathiksha@kali)~[/permission_test]
$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied

(prathiksha@kali)~[/permission_test]
$ cd sample_dir

(prathiksha@kali)~[/permission_test/sample_dir]
$ cat inner.txt
inner content

(prathiksha@kali)~[/permission_test/sample_dir]
$ cd ..
```

Read and execute were denied on the file. Listing the directory was denied, but entering it succeeded - and a known file inside could still be read.

Testing 500 (r-x)

```
(prathiksha@kali)~[/permission_test]
$ chmod 700 sample.txt sample_dir

(prathiksha@kali)~[/permission_test]
$ chmod 500 sample.txt sample_dir

(prathiksha@kali)~[/permission_test]
$ ls -l
total 8
dr-x----- 2 prathiksha prathiksha 4096 Aug 22 18:11 sample_dir
-r-x----- 1 prathiksha prathiksha 18 Aug 22 18:16 sample.txt

(prathiksha@kali)~[/permission_test]
$ cd sample.txt
cd: not a directory: sample.txt

(prathiksha@kali)~[/permission_test]
$ echo "test3" >> sample.txt
zsh: permission denied: sample.txt

(prathiksha@kali)~[/permission_test]
$ ls sample_dir
inner.txt

(prathiksha@kali)~[/permission_test]
$ cd sample_dir

(prathiksha@kali)~[/permission_test/sample_dir]
$ cd ..
```

Read succeeded, modify was denied on the file. Both listing and entering the directory succeeded.

Testing 600 (rw-)

```
(prathiksha@kali)~[/permission_test]
$ chmod 700 sample.txt sample_dir

(prathiksha@kali)~[/permission_test]
$ chmod 600 sample.txt sample_dir

(prathiksha@kali)~[/permission_test]
$ ls -l
total 8
drw----- 2 prathiksha prathiksha 4096 Aug 22 18:11 sample_dir
-rw----- 1 prathiksha prathiksha 18 Aug 22 18:16 sample.txt

(prathiksha@kali)~[/permission_test]
$ cat sample.txt
hello world
test2

(prathiksha@kali)~[/permission_test]
$ echo "test4" >> sample.txt

(prathiksha@kali)~[/permission_test]
$ ls sample_dir
ls: cannot access 'sample_dir/inner.txt': Permission denied
inner.txt

(prathiksha@kali)~[/permission_test]
$ cd sample_dir
cd: permission denied: sample_dir

(prathiksha@kali)~[/permission_test]
$ touch sample_dir/file2.txt
touch: cannot touch 'sample_dir/file2.txt': Permission denied
```

Read and modify both succeeded on the file. Listing the directory succeeded, but entering and creating a file were denied.

Testing 700 (rwx)

```
(prathiksha@kali)~/permission_test
$ chmod 700 sample.txt sample_dir
(prathiksha@kali)~/permission_test
$ chmod 700 sample.txt sample_dir
(prathiksha@kali)~/permission_test
$ ls -l
total 8
drwx----- 2 prathiksha prathiksha 4096 Aug 22 18:11 sample_dir
-rwx----- 1 prathiksha prathiksha  24 Aug 22 18:24 sample.txt
(prathiksha@kali)~/permission_test
$ cd sample.txt
cd: not a directory: sample.txt
(prathiksha@kali)~/permission_test
$ echo "test5" >> sample.txt
(prathiksha@kali)~/permission_test
$ ls sample_dir
inner.txt
(prathiksha@kali)~/permission_test
$ cd sample_dir
(prathiksha@kali)~/permission_test/sample_dir
$ touch newfile2.txt
(prathiksha@kali)~/permission_test/sample_dir
$ rm newfile2.txt
(prathiksha@kali)~/permission_test/sample_dir
$ cd ..
```

Read, modify were successful on the file. Listing, entering, creating, and deleting in the directory all succeeded.

Permission	Regular file test	Observation	Directory test	Observation
400 (r--)	Read / modify	Read succeeds modify fails	List / Enter	List Succeeds Enter fails
200 (-w-)	Read / modify	Read fails Modify succeeds	List / Enter / create	All fail
100 (--x)	Read / execute	Read fails execute fails	List / Enter	List fails Enter succeeds
500(r-x)	Read / Modify	Read succeeds Modify fails	List / Enter	Both succeed
600(rw-)	Read / Modify	Both succeed	List / Enter / create	List succeeds Enter / create fail
700(rwx)	Read / Modify / Execute	All succeed	List / Enter / create / delete	All succeed

Questions and Answers

1. What does read (r) permission allow you to do with a regular file?

It lets you view the file's contents — for example using `cat` or opening it in an editor. It does not allow you to modify or execute the file.

2. What does read (r) permission allow you to do with a directory?

It lets you list the names of the files and subdirectories inside it (using `ls`). It does not allow you to enter the directory or access detailed file information — that requires execute permission as well.

3. What does write (w) permission mean for a regular file?

It means you can modify, append to, or overwrite the file's content. It does not allow you to read the file.

4. What does write (w) permission mean for a directory?

It means you can create, delete, or rename entries (files/folders) inside that directory. This is controlled by the directory's permission, not the permission of the individual files inside it - for example, deleting a file depends on write permission on the directory, not on the file itself.

5. What does execute (x) permission mean for a regular file?

It means you have permission to run the file as a program or script. However, this only works if the file actually contains valid executable content (a plain text file will still fail to run, even with execute permission).

6. Why is execute (x) permission important for accessing a directory?

Execute permission on a directory works as "search" or "traversal" permission. It allows you to `cd` into the directory and access files inside it by name — even without read (list) permission. Without execute permission, you cannot enter the directory or interact with anything inside it, regardless of other permissions.

7. Which permission combination allows the owner to fully access a regular file?

`rw-` (600) provides full access to the file's data — you can read and modify it. `rwX` (700) is needed additionally if the file is also meant to be executed as a program or script.

8. Which permission combination allows the owner to fully access a directory?

`rwX` (700) — read permission to list the contents, execute permission to enter/traverse the directory, and write permission to create, delete, or rename files inside it.

9. Based on your observations, explain why the same `r`, `w`, and `x` permissions behave differently for a regular file and a directory.

A directory is fundamentally different from a regular file — it doesn't store data directly, but instead stores a table of filenames mapped to their inodes. Because of this, the same three permission bits are interpreted differently:

- Read (`r`): On a file, it allows reading the actual content. On a directory, it allows reading the list of filenames stored in it (not their content).
- Write (`w`): On a file, it allows changing the file's content. On a directory, it allows adding, removing, or renaming entries in that filename table.
- Execute (`x`): On a file, it allows running it as a program. On a directory, it allows traversing into it and accessing entries by name — which is a prerequisite for almost any other operation inside that directory.